Simple Inequalities

(Also available in Pyret)

Students identify solutions and non-solutions of inequalities using an interactive starter file. This lesson also reviews the `Boolean` datatype.

Lesson Goals	Students will be able to: • Describe the solution set of a simple inequality
Student-Facing Lesson Goals	• I can find solutions and non-solutions to an inequality
Prerequisites	• Simple Data Types • Contracts • Functions: Contracts, Examples & Definitions • Solving Word Problems with the Design Recipe
Materials	• Lesson Slides • Boolean Starter File • Simple Inequalities Starter File • Boolean Functions (Page 26) • Simple Inequalities (Page 27) • *Optional: Word Problem: hot?* • **Classroom visual:** Language Table
Supplemental Resources	• Booleans Review
Key Points for the Facilitator	• A ***Boolean*** is just another ***data type***, like Numbers or Images. But unlike those types, there are only *two* values: `true` and `false`. While simple to explain, this different behavior can be confusing for some students. • Boolean-producing functions are essentially yes-or-no questions, so the names of the functions in this lesson read like questions. • For example, `odd?` and `even?` (pronounced "odd-huh?" and "even-huh?" in Scheme) are both functions that ask if a given number is odd or even. }

Glossary

Boolean :: a type of data with two values: true and false

data type :: a way of classifying values, such as: Number, String, Image, Boolean, or any user-defined data structure

equation :: a mathematical description of the relationship between two variables or quantities, in which they are equal

inequality :: a mathematical description of the relationship between two variables or quantities, in which they are not necessarily equal

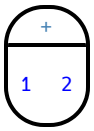
solution set :: the set of all values or coordinate pairs that will make an equation, inequality, set of equations or set of inequalities true

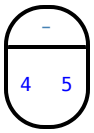
Introducing Booleans *20 minutes*

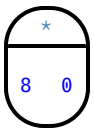
Launch

Students should be logged into WeScheme.

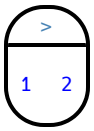
Ask students to evaluate Circles of Evaluation for simple expressions they've seen before, and ask them to convert them into code.

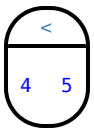
- 

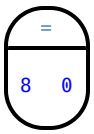
- 

- 

Then show them unfamiliar Circles of Evaluation, and ask them to hypothesize what they think they mean, what they will evaluate to, and what the code would look like.

- 

- 

- 

Have students convert these Circles to code and type them in. What did they evaluate to? What do they think the outputs mean?

Values like `true` and `false` obviously aren't Numbers or Images. But they also aren't Strings, or else they would have quotes around them. We've found a *new data type* , called a ***Boolean***.

Investigate

Have students open the Boolean Starter File.

- Explore the five functions in this starter file: `odd?` , `even?` , `less-than-one?` , `continent?` , and `primary-color?`
- All five functions produce ***Booleans***. Through your exploration, see if you can come up with an explanation of what a ***Boolean*** is.

A ***Boolean*** is just another ***data type***, like Numbers or Images. But unlike the others there are only two values: `true` and `false`.

- Turn to Boolean Functions and use the starter file to complete the questions, identifying inputs that will make each function produce `true`, and inputs that will make each function `false`.

Common Misconceptions

- Many students - especially traditionally high-achieving ones - will be very concerned about writing examples that are "wrong." The misconception here is that an expression that produces `false` is somehow *incorrect*. You can preempt this by explaining that our Boolean-producing functions *should sometimes return false*.

Synthesize

- Students will see functions on this page that they've never encountered before! But instead of answering their questions, encourage them to make a *guess* about what they do, and then type it in to discover for themselves.
- Explicitly point out that *everything they know still works!* They can use their reasoning about Circles of Evaluation and Contracts to figure things out.

Introducing Inequalities *20 minutes*

Overview

Students discover (or expand their understanding of) inequalities by identifying solutions and non-solutions and connecting expressions to graphic representations.

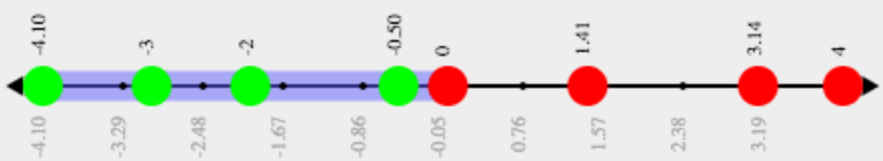
Launch

How are equations different from inequalities?

Equations typically have finite solution sets: there's only one answer for an unknown, or perhaps several answers. ***Inequalities***, on the other hand, can have *infinite* solutions. Inequality expressions divide all of the numbers in the universe into two categories: solutions and non-solutions.

It is important that students learn to recognize that there are many possible solutions and non-solutions to an inequality and are able to identify whether or not a given number is or isn't part of the ***solution set****.*

- We are going to practice identifying whether or not a given number is part of the ***solution set***.
- Open the Simple Inequalities Starter File and click "Run".
- Analyze the graph that appears (image below), as well as the provided code (lines 10, 18, and 26).



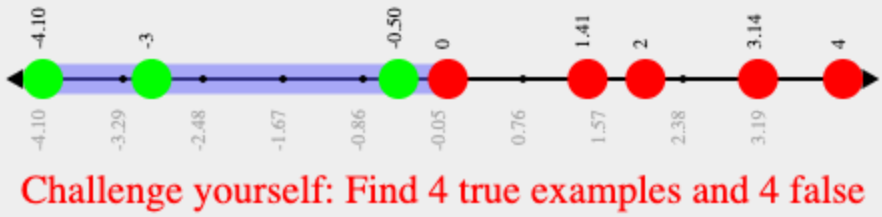
- What do you Notice? What do you Wonder?

Students might observe the following:

- This starter file includes a special `inequality` function that takes in a function *(which tests numbers in an inequality)* and a list of 8 numbers *(to test in the function)*.
- When we click "Run", we see a graph of the inequality on a number line.
- The solution set is shaded in blue.
- The 8 numbers provided in the list are shown as dots on the numberline. They will appear:
  - green *when they're part of the solution set*
  - red *when they are non-solutions*

- Look at line 18. Edit the list of values by deleting *one* of the `-` symbols.

- Hit "Run". Examine the graph that appears (sample image below).



Challenge yourself: Find 4 true examples and 4 false

- How is this graph different from the one you first produced?

A successful input in this starter file will include 4 solutions and 4 non-solutions; in other words, the image returned will show 4 green dots and 4 red dots.
When students modify the list of numbers, they will see there are now 5 red dots and 4 green dots - along with a message that says, *"Challenge yourself: Find 4 true examples and 4 false"* .

Investigate

- Open to the Simple Inequalities and complete it with a partner, identifying solutions and non-solutions to each inequality and testing them in the Simple Inequalities Starter File.
- For each inequality, you must find four solutions and four non-solutions.
- Try using negatives, positives, fractions and decimals as you generate your lists.

Synthesize

- What patterns did you observe in how the inequalities worked?

Additional Exercises:

- Word Problem: hot?